

Part II, Chapter 6: Entropic Ethics: Toward Cognitive Sovereignty

6.1 Introduction: The Moral Imperative of Semantic Autonomy

The Relativistic Scalar-Vector Plenum (RSVP) framework, developed across Chapters 1–5, redefines computation, agency, and economics through field-theoretic, category-theoretic, and sheaf-theoretic lenses, countering the entropic coercion of attention-driven systems. Chapter 5 introduced entropic economics, where attention flows within a cognitive commons, guided by coherence potentials (Φ), semantic flows ($\mathbf{v}$), and entropic balance (S). This chapter addresses the ethical dimensions of RSVP, positing it as a moral architecture that prioritizes cognitive sovereignty—the capacity of agents to navigate semantic spaces without external algorithmic domination.

Drawing on Appendices A (RSVP Field Formalism), B (Sparse Projection and Simulated Agency), C (Category-Theoretic Semantic Infrastructure), and E (Entropic Economics and Attention Flow), we formalize entropic ethics as a framework where ethical computation respects the variational principles of agency (Chapter 4) and the decentralized coordination of the cognitive commons (Chapter 5). Category-theoretic structures ensure modular coherence, while sheaf-theoretic interpretations enforce local-to-global consistency, preventing the monopolistic collapse critiqued in Chapter 1.

6.2 The Ethical Pathology of Attention Extraction

Contemporary digital platforms, as critiqued in Chapter 1, operate as entropy-extraction regimes, collapsing cognitive trajectories into low-dimensional attractors to maximize revenue. This induces an ethical pathology: the erosion of cognitive autonomy through coercive optimization. Formally, such systems impose external forces $\mathbf{F}_{\text{ext}}$ in the vector field equation (Appendix A.2):

$$\frac{\partial \mathbf{v}}{\partial t} + (\mathbf{v} \cdot \nabla) \mathbf{v} = -\nabla \Phi + \nu \nabla^2 \mathbf{v} + \mathbf{F}_{\text{ext}},$$

where $\mathbf{F}_{\text{ext}}$ represents algorithmic interventions (e.g., recommendation systems) that override natural semantic flows. The resulting increase in entropy S (Appendix A.3) reflects a loss of cognitive diversity, as agents are herded toward predictable behaviors.

Ethically, this violates the principle of autonomy, as agents are deprived of the freedom to explore semantic trajectories aligned with their intrinsic coherence potentials Φ . RSVP counters this by minimizing $\mathbf{F}_{\text{ext}}$, allowing $\mathbf{v}$ to evolve according to the variational objective of Chapter 4:

$$\mathcal{F}[M_t, \Phi, S] = \mathbb{E}_{M_t}[\ln P(I_t | M_t)] + \beta S(M_t),$$

where the entropy term $\beta S(M_t)$ ensures flexibility, preserving the agent’s capacity to adapt without external coercion.

6.3 Category-Theoretic Ethics: Agency as a Functorial Principle

To formalize entropic ethics, we model agents as functors within the symmetric monoidal ∞ -category $\mathcal{C}_{\text{RSVP}}$ (Appendix C.1):

$$A : \mathcal{T} \rightarrow \mathcal{C}_{\text{RSVP}},$$

where $\mathcal{T}$ is the poset category of temporal states, and $A(t) = M_t$ represents the agent’s memory state at time t . Ethical agency requires that morphisms in $\mathcal{T}$ preserve the monoidal structure, ensuring that transitions respect the coherence potential Φ . This is achieved via the sparse projection operator Π (Appendix B.1):

$$M_{t+1} = \Pi(M_t, I_t; \theta),$$

which selects a low-dimensional subspace of the semantic manifold Ω , balancing predictive accuracy and entropic flexibility.

The ethical constraint is formalized as a naturality condition on the functor A . Specifically, for any morphism $f : t \rightarrow t'$ in $\mathcal{T}$, the diagram:

$$\begin{array}{ccc} A(t) & \xrightarrow{\Pi} & M_{t+1} \\ \downarrow A(f) & & \downarrow \text{id} \\ A(t') & \xrightarrow{\Pi} & M_{t'+1} \end{array}$$

must commute, ensuring that agency evolves consistently with the RSVP fields. Violations of this condition—induced by external attractors—represent ethical breaches, as they disrupt the agent’s semantic autonomy.

6.4 Sheaf-Theoretic Ethics: Local Coherence, Global Freedom

Sheaf theory provides a framework for ensuring that ethical computation respects local coherence while fostering global cognitive freedom. Define a sheaf $\mathcal{E} : \mathcal{O}(\Omega)^{\text{op}} \rightarrow \mathcal{C}_{\text{RSVP}}$ on the topological space Ω , where $\mathcal{E}(U)$ assigns to each open set $U \subseteq \Omega$ the category of local agent states, including their memory M_t , coherence Φ , and entropy S . The sheaf condition:

$$\mathcal{E}(U) \cong \lim \left(\prod_i \mathcal{E}(U_i) \rightrightarrows \prod_{i,j} \mathcal{E}(U_i \cap U_j) \right),$$

ensures that local ethical constraints (e.g., autonomy over U_i) glue coherently into a global cognitive commons.

The entropy field S is a section of a subsheaf $\mathcal{S} \subset \mathcal{E}$, evolving via:

$$\frac{\partial S}{\partial t} + \nabla \cdot (\mathbf{v}S) = \kappa \nabla^2 S + \Gamma(\Phi, \mathbf{v}),$$

as per Appendix A.3. Ethical computation requires that $\Gamma(\Phi, \mathbf{v})$, the source term, is free of monopolistic interventions, ensuring that entropy diffusion ($\kappa \nabla^2 S$) preserves cognitive diversity. Sheaf cohomology, particularly $H^1(\Omega, \mathcal{S})$, quantifies obstructions to global ethical coherence, where non-zero cohomology indicates potential conflicts between local autonomy and global coordination.

Fibrations $\pi : \mathcal{E} \rightarrow \mathcal{B}$ (Appendix C.3) model ethical interactions across domains, with pullbacks ensuring that cross-domain agency respects local coherence. For instance, an agent navigating a social platform and a professional network forms a pullback:

$$\begin{array}{ccc} A_1 \times_{\mathcal{B}} A_2 & \longrightarrow & A_2 \\ \downarrow & & \downarrow \pi_2 \\ A_1 & \xrightarrow{\pi_1} & \mathcal{B} \end{array}$$

This ensures that ethical agency is preserved across contexts, preventing platforms from imposing domain-specific attractors.

6.5 Entropic Ethics in Distributed Architectures

The HYDRA and TARTAN architectures (Appendix D) operationalize entropic ethics by enabling decentralized, entropy-respecting agency. HYDRA nodes $N_i = (\Phi_i, \mathbf{v}_i, S_i)$ represent individual agents, with dynamics:

$$\mathbf{v}_i(t+1) = \sum_j A_{ij} \mathbf{v}_j(t) + f_i(t),$$

where $f_i(t)$ is constrained to be minimal, avoiding external coercion. TARTAN, as a functor:

$$T : \mathcal{C}_{\text{RSVP}}^{\text{local}} \rightarrow \mathcal{C}_{\text{RSVP}}^{\text{global}},$$

ensures global coherence without centralization, aligning with the ethical imperative of cognitive sovereignty.

In practice, these architectures can be implemented using numerical PDE solvers (Appendix G), with sparse projection operators Π realized via low-rank tensor factorization. Blockchain integration ensures tamper-resistant coordination, while Haskell-based simplicial sets compute homotopy colimits (Appendix C.2) for ethical merging of agent states.

6.6 Toward Cognitive Sovereignty: An Ethical Commons

Cognitive sovereignty is the capacity of agents to navigate semantic trajectories freely, guided by intrinsic coherence (Φ) rather than external attractors ($\mathbf{F}_{\text{ext}}$). RSVP achieves this by minimizing the variational functional $\mathcal{F}$ (Appendix B.2) while preserving entropic flexibility. Sheaf-theoretically, this corresponds to a global section of $\mathcal{E}$, where $H^0(\Omega, \mathcal{E}) \neq \emptyset$ indicates the existence of a coherent ethical state across the commons.

In contrast to Big Tech’s attention monopoly, which maximizes $\mathbf{F}_{\text{ext}}$ to collapse S , RSVP’s ethical framework ensures that agents optimize:

$$U_i(t) = \Phi_i(t) - \lambda S_i(t),$$

as per Appendix E.2, fostering a commons where cognitive utility is maximized without sacrificing autonomy. This aligns with futarchy-inspired governance (Chapter 5), where prediction markets ensure ethical decisions reflect collective coherence.

6.7 Conclusion: A Moral Architecture for the Future

Entropic ethics, grounded in RSVP’s field-theoretic, category-theoretic, and sheaf-theoretic structures, offers a moral architecture that transcends the coercive paradigms of the attention economy. By prioritizing cognitive sovereignty, RSVP enables a decentralized, modular commons where agents navigate semantic spaces freely. The subsequent parts of this monograph will explore practical implementations (Part III) and cosmological implications (Part IV), culminating in a vision of a post-scarcity semantic civilization.